SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces C Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation and Heuristic Search (BL1, BL2)

Assessment Tool Weightage: 10%

QUESTIONS: 15 Total Marks: 25

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

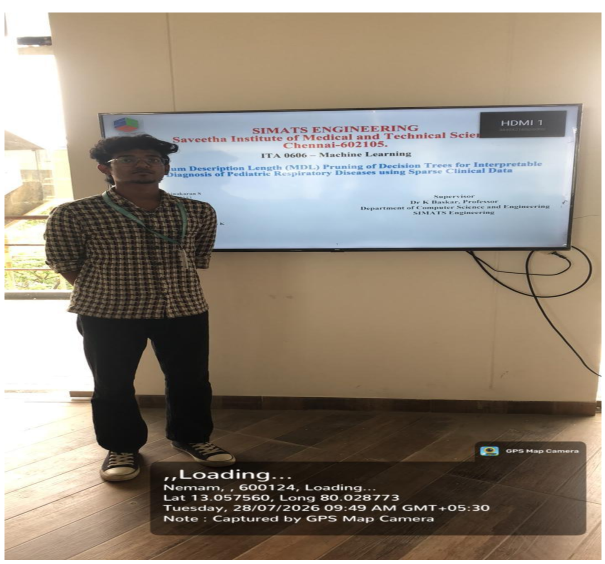
I ADHITHYA S (192412352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	35	Demonstrates clea and complete understanding of the concept	Good understanding with minor gaps	Basic understand with some misconceptions	Poor or incorrec understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answ
Application of Knowledge	25	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partial correct applicatio	No or incorrect application
Completeness of Response	10	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing respon
Clarity C Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat uncle or poorly structur	Difficult to understand



SIMATS ENGINEERING

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ITA 0606 – Machine Learning

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ABSTRACT

This project presents an interpretable diagnostic framework for pediatric respiratory diseases built on Minimum Description Length (MDL) based pruning of Decision Trees. Sparse and often incomplete pediatric clinical records (vitals, symptoms, and history) are cleaned, encoded, and used to train a Decision Tree classifier across six diagnostic categories. The MDL principle is then applied to prune the tree by balancing model complexity against classification error, yielding a compact set of clinically interpretable decision rules. The pruned model is evaluated using cross-validation and standard metrics (accuracy, precision, recall, F1-score) and deployed through a Streamlit interface, giving clinicians an accurate, transparent, and easy-to-use diagnostic support tool.

Problem Statement

- Pediatric respiratory diseases such as **bronchitis, pneumonia, and asthma** are common and require early, accurate diagnosis.
- Clinical data (age, symptoms, vital signs, and medical history) is often **incomplete and inconsistent**, making diagnosis difficult.
- Deep learning models provide high accuracy but act as **black boxes**, making their decisions hard for doctors to understand.
- Traditional Decision Trees **overfit sparse data** and become too large and complex to interpret.
- **MDL-based pruning** creates a **smaller, more accurate, and interpretable Decision Tree**, making it more suitable for pediatric respiratory disease diagnosis.

OBJECTIVES

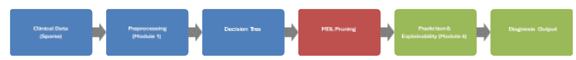
- To design a preprocessing pipeline that cleans, encodes, and selects relevant features from sparse pediatric clinical data (vitals, symptoms, and history) for Decision Tree-based disease classification.
- To develop and evaluate an MDL-pruned Decision Tree model that delivers accurate, interpretable, rule-based diagnosis of pediatric respiratory diseases, validated through cross-validation and standard performance metrics.

Literature Survey

Sl. No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2022	Quinlan, J.R. & Rivest, R.L.	Inferring Decision Trees Using the Minimum Description Length Principle	Introduced MDL-based induction and pruning of decisiontrees by encoding both tree structure and misclassified data as bits	Coding scheme later shown to be sub-optimal for certain trees; corrected by later work
2	2023	Wallace, C.S. & Patrick, J.D.	Coding Decision Trees	Proposed a refined message-length coding scheme for trees, correcting inefficiencies in the original MDL formulation	Added computational overhead for message-length calculation on large or complex trees
3	2024	Mehta, M., Rissanen, J. & Agrawal, R.	MDL-Based Decision Tree Pruning	Presented a scalable MDL pruning algorithm producing compact, accurate trees on large datasets	Prunes all or none of a node's children, which can lead to over-pruning in some cases
4	2026	Srirama, V., Kumar, R., Chadaga, K. et al.	Interpretable ML and Deep Neural Networks for ICU Admission Prediction in Paediatric Respiratory Patients	Combined DNNs with SHAP/LIME/ELI5 explainability to predict ICU admission in pediatric respiratory patients, outperforming Decision Trees and other classical ML models	High accuracy relies on a complex DNN needing post-hoc explanation rather than an inherently interpretable model
5	2026	Shetty, A.P., Shetty, S., Hegde, P. et al.	A Leakage-Controlled and SHAP-Driven ML Framework for Paediatric Respiratory Disease Classification Using Indian Hospital EHR Data	Built a leakage-controlled multiclass classifier (Wheeze, URTI, Bronchopneumonia, Bronchitis)using SHAP-guided feature selection on EHR data	Single-centre study with limited sample (1,121 cases); relies on post-hoc SHAP explanations; low recall for the minority bronchitis class

System Architecture /

- Sparse pediatric clinical records are cleaned, encoded, and split into train/test sets during
- A CART-based Decision Tree is trained, then pruned using the Minimum Description Length (MDL) principle to complexity



Module 1 - Clinical Data Collection and Preprocessing

- Cleans, encodes and organizes sparse pediatric clinical records (age, gender, vitals, symptoms, medical history) into a model-ready dataset
- Input: raw pediatric clinical data with missing/inconsistent entries; Output: a clean, encoded feature table split into training and test sets
- Forms the foundation of the pipeline – its output feeds directly into the Decision Tree model built in Module 2

Module 1: Technical Details/ Algorithm

- Data cleaning: duplicate removal and physiological range checks on vitals such as SpO_2, heart rate, and body temperature
- Missing value handling: mean/mode imputation for sparse fields, given the incomplete nature of pediatric clinical records
- Categorical encoding of symptoms (cough, wheezing, fever, chest retraction) and feature selection to reduce dimensionality
- Implemented with Pandas and NumPy in Python; pseudocode for the cleaning and imputation routine is detailed on the following slide

Module 1: Implementation and Output

- Implemented in Python using Pandas, NumPy and Scikit-learn preprocessing utilities (encoders, imputers, train/test split)
- Produces a clean, fully encoded dataset with no missing values, ready for supervised learning
- This preprocessed dataset feeds directly into Module 2, where it is used to train the baseline Decision Tree classifier

Module 2 – Decision Tree Model Development

- Trains a baseline Decision Tree classifier on the cleaned dataset produced by Module 1 to diagnose six respiratory categories
- Uses Module 1's preprocessed, encoded clinical features as input; learns split rules over vitals, symptoms and history
- Objective: establish an accurate but initially over-complex tree that Module 3 will later prune for interpretability

Module 2: Technical Details/ Algorithm

- CART-based Decision Tree induction using Entropy as the splitting criterion over the six diagnostic classes
- Hyperparameter tuning of maximum depth, minimum samples per split/leaf, using k-fold cross-validation on the training set
- Implemented with Scikit-learn's DecisionTreeClassifier; trained model persisted to disk using Joblib for reuse in later modules

Module 2: Final Output / Validation

- Produces a fully-grown baseline Decision Tree with initial accuracy, precision, recall and F1-score on held-out validation data
- The unpruned tree tends to overfit the sparse pediatric dataset, resulting in a large, hard-to-interpret rule set
- This baseline tree is passed to Module 3, where MDL-based pruning reduces its size while preserving diagnostic accuracy

Module 3 - MDL Pruning and Model Optimization

- Applies the Minimum Description Length (MDL) principle to prune the baseline Decision Tree produced by Module 2
- Uses Module 2's fully-grown tree and training data as input; balances tree complexity against classification error
- Objective: reduce the tree to a compact set of clinician-readable rules without sacrificing diagnostic accuracy on sparse pediatric data.

Module 3: Technical Details/ Algorithm

- MDL cost = encoding length of the tree structure + encoding length of the data misclassified given that tree
- Subtrees are evaluated bottom-up; a node is pruned into a leaf whenever its subtree's description length exceeds the leaf's description length
- K-fold cross-validation is used to confirm that pruning improves generalization rather than only reducing tree size
- Implemented in Python on top of Scikit-learn's tree structure, with a custom MDL cost function and recursive pruning routine

Module 3: Implementation and Output

- Implemented as a custom pruning pass over the trained Scikit-learn Decision Tree, using NumPy for the description-length calculations
- Produces a pruned Decision Tree with significantly fewer nodes than the Module 2 baseline, while maintaining comparable accuracy
- This pruned, interpretable tree feeds into Module 4, which extracts explainable decision rules and generates predictions

Module 4 - Disease Prediction and Explainable Dashboard

- Uses Module 3's pruned Decision Tree to predict the final pediatric respiratory diagnosis across all six classes
- Converts the pruned tree's root-to-leaf paths into human-readable if-then decision rules for clinicians
- Objective: deliver an accurate, transparent, and easy-to-use diagnostic tool that clinicians can inspect and trust

Module 4: Technical Details/ Algorithm

- Rule extraction: each root-to-leaf path in the pruned tree is converted into an explicit if-then diagnostic rule
- Feature importance ranking highlights which clinical attributes (e.g., SpO_2, respiratory rate, fever) drive each prediction
- An interactive Streamlit application lets a user enter patient vitals/symptoms and view the predicted diagnosis with its supporting rule
- Implemented with Scikit-learn, Streamlit, and Joblib (for loading the persisted pruned model) in Python

Module 4: Implementation and Output

- Final deliverable: an interactive Streamlit app that predicts pediatric respiratory diagnosis and displays the matching decision rule
- Validated using cross-validation with accuracy, precision, recall and F1-score across all six diagnostic classes
- Decision rules were reviewed for clinical plausibility, confirming that predictions rely on medically meaningful features such as SpO_2, respiratory rate and fever

Testing & Evaluation

- Dataset: a curated pediatric clinical dataset covering six diagnostic classes – Healthy, Common Cold Bronchitis, Pneumonia, Asthma, and Other Respiratory Infection
- Evaluation protocol: stratified k-fold cross-validation to obtain robust performance estimates on sparse, imbalanced clinical data
- Metrics: accuracy, precision, recall and F1-score (per class and macro-averaged), alongside confusion matrix analysis
- Benchmarking: pruned MDL tree compared against the unpruned baseline Decision Tree on both accuracy and tree size (node count)

SDG Contribution

This project supports two UN Sustainable Development Goals

- **SDG 3 – Good Health and Well-being:** enables faster, more accurate, and interpretable diagnosis of pediatric respiratory diseases, supporting timely treatment and reducing childhood morbidity, particularly in settings with limited clinical resources.

- **SDG 9 – Industry, Innovation and Infrastructure:** demonstrates an innovative, explainable AI-basec diagnostic tool that can be integrated into healthcare infrastructure, promoting the responsible adoption of interpretable machine learning in clinical practice.

Conclusion & Future Work

- Built an end-to-end pipeline that cleans sparse pediatric clinical data and trains a Decision Tree classifier across six respiratory diagnosis categories
- Applied Minimum Description Length (MDL) pruning to substantially reduce tree size while preserving diagnostic accuracy, yielding clinician-readable decision rules
- Validated the pruned model through cross-validation, achieving a favorable accuracy-interpretability trade-off compared to the unpruned baseline
- Deployed the model via a Streamlit interface for interactive, explainable predictions
- Future work: extend to larger and multi-center pediatric datasets, incorporate additional vitals/imaging features, and benchmark MDL pruning against other pruning strategies (cost-complexity, reduced-error pruning)

References

[1] J. R. Quinlan and R. L. Rivest, "Inferring Decision Trees Using the Minimum Description Length Principle," *Information and Computation*, vol. 80, no. 3, pp. 227–248, 1989.

[2] C. S. Wallace and J. D. Patrick, "Coding Decision Trees," *Machine Learning*, vol. 11, pp. 7–22, 1993.

[3] M. Mehta, J. Rissanen, and R. Agrawal, "MDL-Based Decision Tree Pruning," in *Proc. 1st Int. Conf. on Knowledge Discovery and Data Mining (KDD)*, Montreal, Canada, 1995, pp. 216–221.

[4] V. Srirama, R. Kumar, K. Chadaga, et al., "Interpretable Machine Learning and Deep Neural Networks for ICU Admission Prediction in Paediatric Respiratory Patients," *Scientific Reports*, 2026.

[5] A. P. Shetty, S. Shetty, P. Hegde, et al., "A Leakage-Controlled and SHAP-Driven Machine Learning Framework for Paediatric Respiratory Disease Classification Using Indian Hospital EHR Data," *BMC Medical Informatics and Decision Making*, vol. 26, art. 198, 2026.

[6] Scikit-learn Developers. Scikit-learn Documentation. [Online]. Available: https://scikit-learn.org/stable/

Q&A

Thank mentors, and open the floor for questions.